Amadreza Farahani | ML/AI Engineer

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Skills

Agentic AI LangGraph, LangChain, LangSmith, MCP, multi-agent, agentic RAG (hybrid retrieval, reranking), guardrails
LLM Systems Transformers, HuggingFace, AWS Bedrock, Ollama, vLLM, TorchServe, Milvus, ChromaDB, Pydantic, LoRa, QLoRa
Machine Learning NumPy, Pandas, scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, MLflow, GANs, U-Net, YOLO, ASR
Cloud Technologies AWS (Bedrock, ECS, EKS, S3, SageMaker), Docker
Software Development Python, C/C++, SQL, Bash, Linux, Git, CI/CD, FastAPI, Prometheus, Grafana, Airflow

Professional Experience

AI Engineer
Turin
Zirak
11/2024 – Current

- Fine-tuned Llama 3.2 3B LLM with **LoRA**/**QLoRA** for an email-triage agent that classifies requests vs. non-requests and extracts structured fields via tool calling (LangGraph + **Pydantic** schemas), running fully on-premise along with ChromaDB for local vector storage.
- Developed a production NL-to-SQL agent for **Poste Italiane**'s logistics division using **LangGraph** with **MCP**-exposed database tools, routing tool calls across **PostgreSQL**, MySQL, and MongoDB; FastAPI with **Ollama** for on-prem inference, cutting query authoring time by ~70% for non-technical analysts.
- Led a team of two to ship a cross-platform multi-agent meeting assistant (Electron/React + Python/FastAPI) combining real-time transcription, diarization, speaker verification, and **agentic RAG** with hybrid retrieval and reranking (**LangChain**/LangGraph) on **AWS Bedrock**; **LangSmith** tracing and regression evals, reducing summarization hallucinations by ~25%.
- Deployed an LLM assistant for consultancy and HR workflows, serving **HuggingFace** models via **vLLM**/**TorchServe** behind **FastAPI** with **Bedrock Guardrails** and **LangSmith** for traceability; saved ~50 hours/month of manual document review, enabling the product for additional clients.
- Designed a custom U-Net for multispectral imagery with an **Airflow**-orchestrated ingestion, training and evaluation pipeline on **AWS**, outperforming Random Forest, **XGBoost** and **LightGBM** baselines by ~10% F1 and reducing manual image review by ~80%.
- Built a rail-safety Computer Vision pipeline (**YOLOv8**/**PyTorch**) to detect sign defects and segment/track rails from nadir and frontal views in production.

Visiting Researcher – Master Thesis
Darmstadt
Technische Universität Darmstadt | CROSSING
08/2023 – 11/2024

- Designed a Diffusion **Transformer** for voice conversion (Python, PyTorch) on **HPC** GPUs, improving speaker similarity by ~9% over open-source baselines with fewer sampling steps, enabling real-time conversion.

Associate Generative AI Engineer
Canelli
AROL S.p.a
09/2022 – 09/2023

- Developed a synthetic machinery data generator (TensorFlow + **Flask** + **MongoDB**) and a customer testing dashboard, packaged with **Docker Compose** (client/server/DB) and REST API docs to avoid exposing proprietary data.
- Designed KPIs and Python tests to validate synthetic vs. real machinery signals and API responses, integrating these checks into the team's **CI pipeline** and reducing manual QA effort for customer demos.
- Reviewed over 5,000 lines of production **Python** and ML pipeline code (training, inference, data preprocessing), adding tests and reducing compute cost while making models easier to debug and extend.

Machine Learning Engineer
Tehran
Part AI Research Center
02/2020 – 03/2022

- Refined and fine-tuned an on-device keyword-spotting Android app in **Java**, using hard negative samples to reduce false positives by ~30%.
- Optimized a speaker verification system on **TensorFlow**, fine-tuned on new domains that reduced the Equal Error Rate (EER) by ~2%.
- Engineered **Test-Driven Development TDD** for Persian **ASR** and **NLP** services; wrote unit/integration tests to reach 100% coverage and set up **Jenkins CI/CD** to automate builds, tests, and releases.
- Mentored new team members on clean, production-ready **Python** (structure, testing, logging), improving code quality and safety.
- Implemented **Neural Network** architectures for scalable products, optimizing for both cost-efficiency and operational safety.

Internship Experience

Computer Vision - Bachelor Thesis
Tehran
Megamouj Co. | University of Science and Technology (IUST)
09/2019 – 02/2020

- Automated a traffic-flow monitoring system (**C++**/Python, **OpenCV**, YOLOv3) for vehicle detection, tracking, and flow estimation, reducing manual video review in a national-scale project.
- Engineered a data analysis pipeline using **NumPy** and **Pandas** to structure annotation datasets, utilizing **scikit-learn** to compute performance metrics and validate system reliability.

Education

M.Sc. Computer Engineering - Artificial Intelligence & Data Analytics
Turin
Politecnico di Torino
2022 - 2024

- **Thesis:** RobVC: Robust Zero-Shot Self Supervised Voice Conversion (Funded by CROSSING Co. in Darmstadt) | Blog

B.Sc. Electrical and Computer Engineering - Telecommunications
Tehran
University of Science and Technology (IUST)
2016 – 2021

- **Thesis:** Deep Learning-Based Vehicle Detection and Traffic Flow Analysis (Funded by Megamouj Co.) | Blog